

Shivam Kumar

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PROFESSIONAL SUMMARY

Energy Engineer with expertise in computational fluid dynamics (CFD), redox flow battery experimentation and simulation, and project management. Proven track record in energy system and market modelling, translating market intelligence into strategic inputs, and leading turbine maintenance initiatives to enhance operational efficiency. Skilled in cross-functional coordination, strategic planning, and advanced simulation tools (NX, StarCCM+, MATLAB), with a focus on decarbonization and renewable energy integration.

EDUCATION

Friedrich-Alexander-Universität (FAU) Erlangen-Nürnberg Oct 2023 - Present
Master of Science (M.Sc.), Clean Energy Processes, GPA: 2.3 (German grade) Erlangen, Germany

- Coursework (Specialisation in Energy Technologies): Focused on advanced modules including Pumps and Turbines, Energy Storage Systems, Efficient Heat Transfer, Energy Market Modelling, Fuel Cells and Electrolysers, Thin Film Processing, and Photovoltaics (PV).

BMS Institute of Technology and Management Aug 2018 - Jun 2022
Bachelor of Engineering (B.E.), Mechanical Engineering, GPA: 2 (German grade) Bangalore, India

- Core modules including but not limited to Energy Engineering, Engineering Mathematics, CAD, Material Engineering, Thermodynamics, Manufacturing and Production Engineering, Fluid Mechanics, Fluid Engineering, CAE, Theory of Machines.

EXPERIENCE

Siemens Energy Jan 2026 – Aug 2026
Master's Thesis Student Mülheim & Erlangen, Germany

- Analysis and improvement of electrolyte flow in a redox flow cell: Computational Fluid Dynamics (CFD) and experimental validation in a laboratory scale test rig.
- Data processing and analysis using Python

Working Student

Jan 2025 – Dec 2025

- Portfolio Management: Technology screening, competitor profiling & analysis, market players scouting; Translated market intelligence & analysis, identifying partnership opportunities to expand the LDES portfolio.
- Battery Simulation: Utilized advanced simulation tools (NX and StarCCM+) to analyse electrolyte flow.

Adani Power

Aug 2022 - Aug 2023

Assistant Manager - Operations & Maintenance

Waidhan, M.P., India

- Project Management: Served as the project manager for Turbine Maintenance and Balance of Plant maintenance projects.
- Planning & Scheduling: Planned and executed maintenance strategies, resource allocation, ensuring minimal downtime and cost efficiency.
- Technical Oversight: Led technical compliance, EHS, mitigated risks, and assured delivery quality.
- Managed procurement activities and supplier coordination for critical spares and services, including vendor evaluation, cost negotiations, quality checks, and performance feedback.

SKILLS

Project Management:

Cross-functional project coordination, Strategic planning, Operation & resource management, Stakeholder management

Energy Engineering:

Industrial STG, Maintenance of turbine systems, Supercritical power plants, Renewable energy, Power plant engineering, Redox Flow Batteries, Energy storage & decarbonization

Programming Languages and Data Analysis:

C/C++, Python, SQL, MATLAB

CAD/CAM Software:

Solid Edge, NX CAD, PTC Creo

CAE Software:

Star-CCM+, ANSYS Fluent, ANSYS workbench (FEM/FEA), NX

Productivity Software:

Linux OS, Microsoft Office Suite, Google Suite